SAP ABAP R2R PROJECT

Record-to-Report: Accounting Document Summary Report

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Program Name: Z343_REUSE_ALV_GRID_DISPLAY

Technology: SAP ABAP / SAP GUI

1. Project Overview

This project demonstrates a classical SAP ABAP report for viewing accounting document summaries in an ALV grid. It retrieves accounting document header information and line-item amounts, groups the line items by company code, document number, and fiscal year, and calculates total debit, total credit, and the debit-credit difference.

2. Project Objectives

- Retrieve accounting document data using selection criteria.
- Join accounting document headers and line items from BKPF and BSEG.
- Calculate debit and credit totals for each accounting document.
- Calculate the difference as total debit minus total credit.
- Display the summarized result in an ALV grid using REUSE_ALV_GRID_DISPLAY.
- Validate selection filters, no-data handling, and the amount calculations.

3. Tools and Technologies

- SAP GUI and ABAP Workbench (SE38)
- ABAP report programming
- BKPF – Accounting Document Header table
- BSEG – Accounting Document Segment / line-item table
- Open SQL INNER JOIN
- Internal tables and field symbols
- Selection-screen fields using SELECT-OPTIONS
- Function module REUSE_ALV_GRID_DISPLAY
- SE16N for table-level validation

4. Program Details

Program name: Z343_REUSE_ALV_GRID_DISPLAY

Report type: Classical executable ABAP report

Output: ALV grid summarizing accounting documents

Selection criteria used: Company Code (S_BUKRS), Posting Date (S_BUDAT), and Document Type (S_BLART).

5. Database Tables

Table	Purpose	Example fields used
BKPF	Accounting document header	BUKRS, BELNR, GJAHR, BLART, BUDAT, BLDAT, WAERS
BSEG	Accounting document line items	BUKRS, BELNR, GJAHR, BUZEI, HKONT, SHKZG, DMBTR

6. Selection Screen

The report provides the following selection options so users can restrict the result set:

- S_BUKRS – company code
- S_BUDAT – posting date range
- S_BLART – accounting document type

Example test values used during validation included company code 0123, posting dates 01.01.2020 through 31.12.2026, and document type DR. Use values that exist in the current SAP system when repeating tests.

7. Processing Logic

1. Read accounting document headers from BKPF and corresponding line items from BSEG.
2. Join the tables using company code (BUKRS), document number (BELNR), and fiscal year (GJAHR).
3. Apply the company-code, posting-date, and document-type selection ranges.
4. Loop through the selected line items and group them by company code, document number, and fiscal year.
5. Add DMBTR to debit total when SHKZG is 'S' and to credit total when SHKZG is 'H'.
6. Calculate Difference = Debit Amount − Credit Amount.
7. Display the summarized results using REUSE_ALV_GRID_DISPLAY.

8. ALV Output Fields

Output field	Meaning	Validation
BUKRS	Company code	Matches selection
BELNR	Accounting document number	Matches source document
GJAHR	Fiscal year	Matches source document
BLART	Document type	Matches selected type when specified
WAERS	Document currency	Matches BKPF
DEBIT_AMT	Total debit amount	Sum of relevant debit line items
CREDIT_AMT	Total credit amount	Sum of relevant credit line items
DIFFERENCE	Debit minus credit	Should equal DEBIT_AMT − CREDIT_AMT

9. Testing and Validation

Test	Expected result
Company code filter	Enter company code 0123 and verify all displayed rows belong to 0123.
Posting date filter	Enter a date range and verify each result's posting date is within that range.
Document type filter	Enter DR and verify all results have document type DR.
No-data scenario	Use a company code or filter combination confirmed to return no records; verify the report's no-data message.
Debit/credit totals	Compare a sample document against BSEG line items using SHKZG and DMBTR.
Difference calculation	Verify Difference = Debit Amount − Credit Amount.
Syntax and activation	Run syntax check and activate the report without errors.

Validation note: Complete each test in the SAP system and record the actual outcome. Do not mark a test as passed unless it has been checked. In particular, investigate any row that shows a company code outside the entered selection, and verify blank or zero totals against the source line items.

10. Screenshot Checklist

- Final_ALV_Output.png – final ALV grid with headings and document rows
- No_Data_Message_Test.png – no-data message after a confirmed no-match selection
- Debit_Credit_Validation.png – ALV totals and difference for a sample document
- Posting_Date_Filter_Test.png – results after applying posting-date range
- Document_Type_Filter_Test.png – results after applying document type
- Syntax_Check_Activation.png – successful syntax check and activation
- Source_Code_1.png to Source_Code_4.png – readable source-code sections
- BSEG_Debit_Credit_Verification.png – source line items with BUZEI, SHKZG, and DMBTR

11. Known Checks Before Submission

- Confirm that all rows in the company-code test show the selected company code.
- Check that the ALV groups data by company code, document number, and fiscal year.
- Verify debit and credit totals against BSEG for at least three documents.
- Confirm that no-data handling is triggered only when no matching rows are found.
- Ensure the output columns and amount formatting are readable.

12. Conclusion

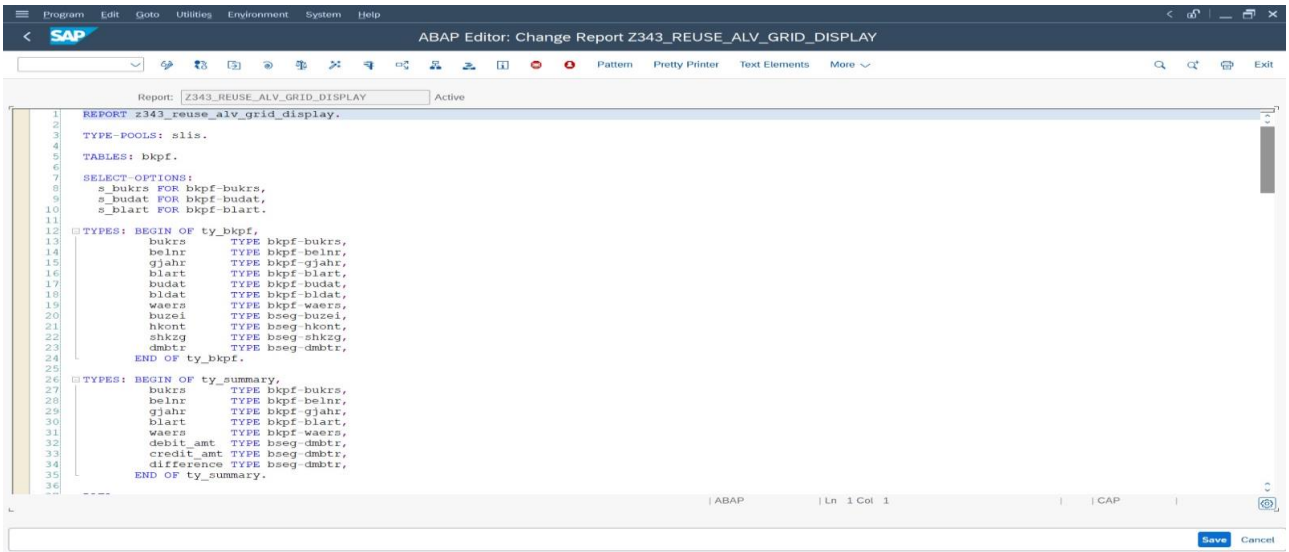
The project demonstrates SAP ABAP reporting with Open SQL joins, selection criteria, internal-table processing, debit and credit aggregation, difference calculation, and ALV display. It provides a summary view of accounting documents and supports basic validation against the source BKPF and BSEG tables. Final completion should be confirmed by running the listed tests and attaching the corresponding screenshots.

Appendix: Screenshot Placement

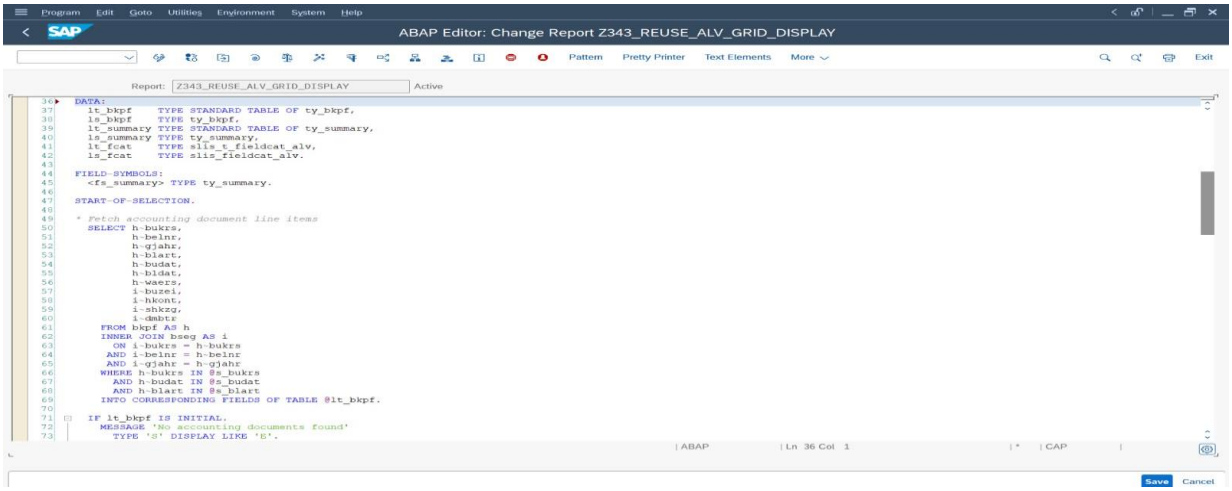
Insert each screenshot under the relevant test section or in this appendix. Add a short caption with the test name, selection values, and actual result. Use only screenshots captured from your own SAP system.

Figure 1: ABAP SOURCE CODE

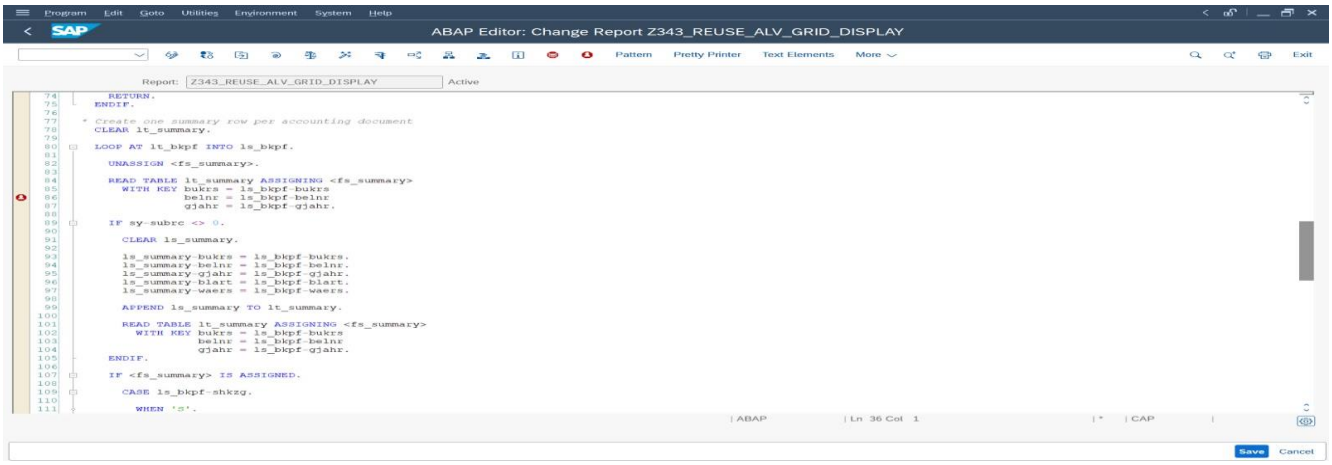
Source_Code_1:



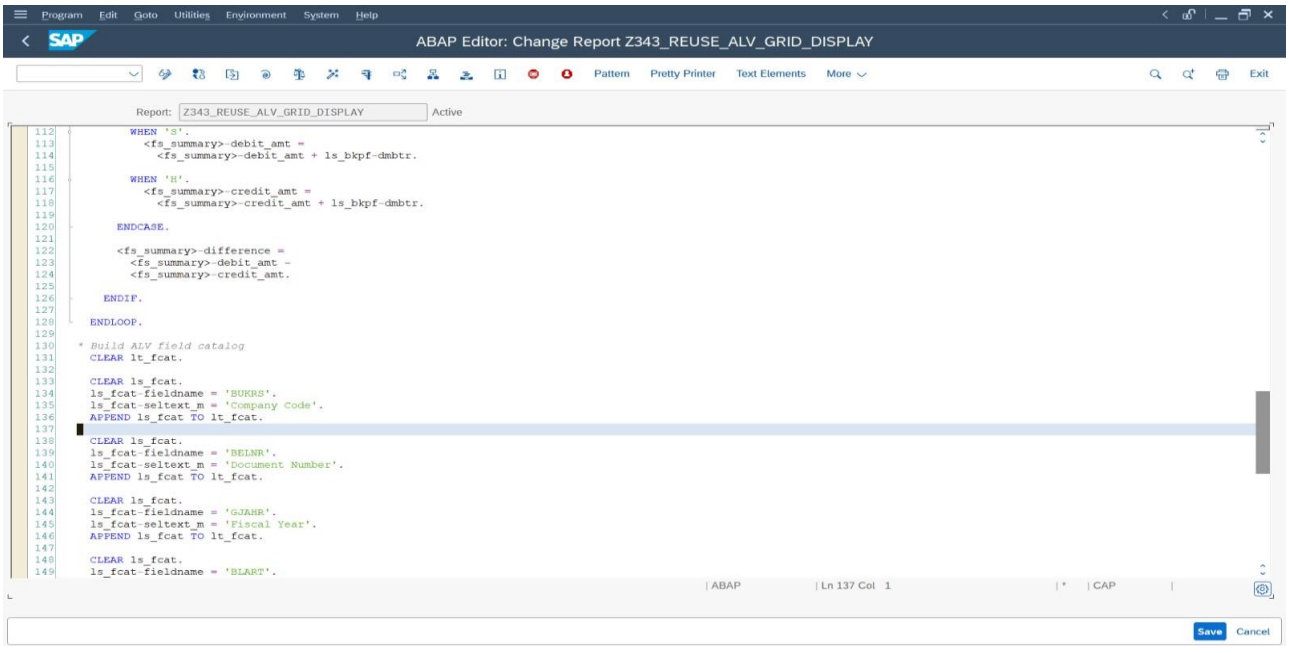
Source_Code_2:



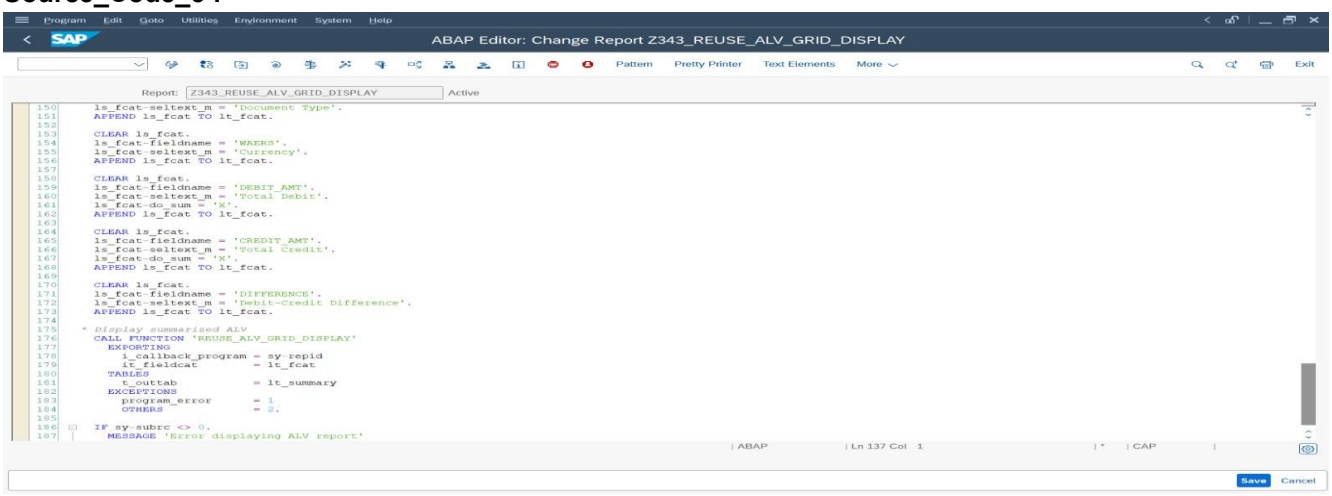
Source_Code_3:



Source_Code_4:



Source_Code_5 :



Source_Code_6:

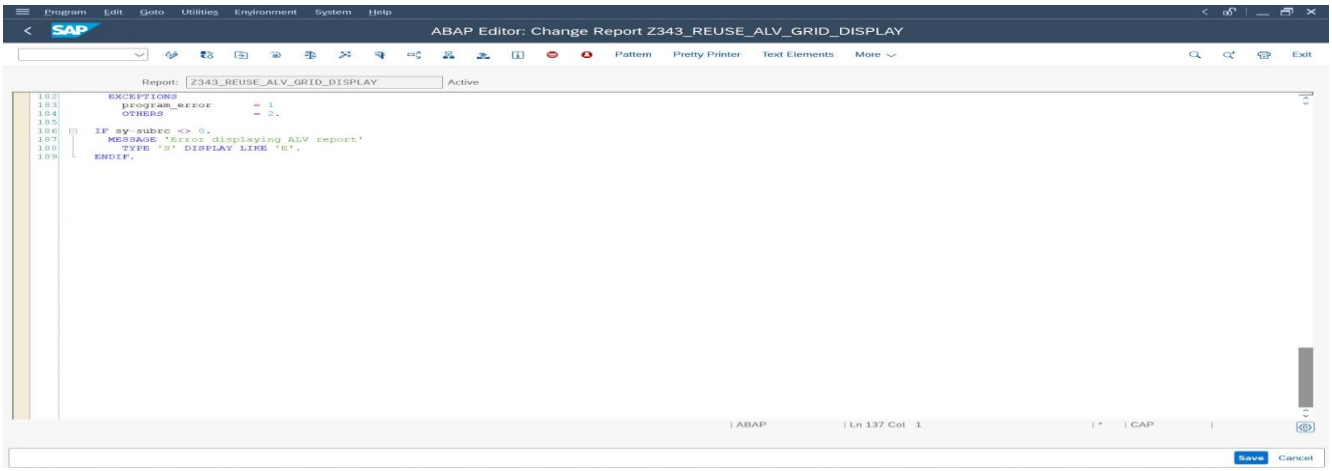
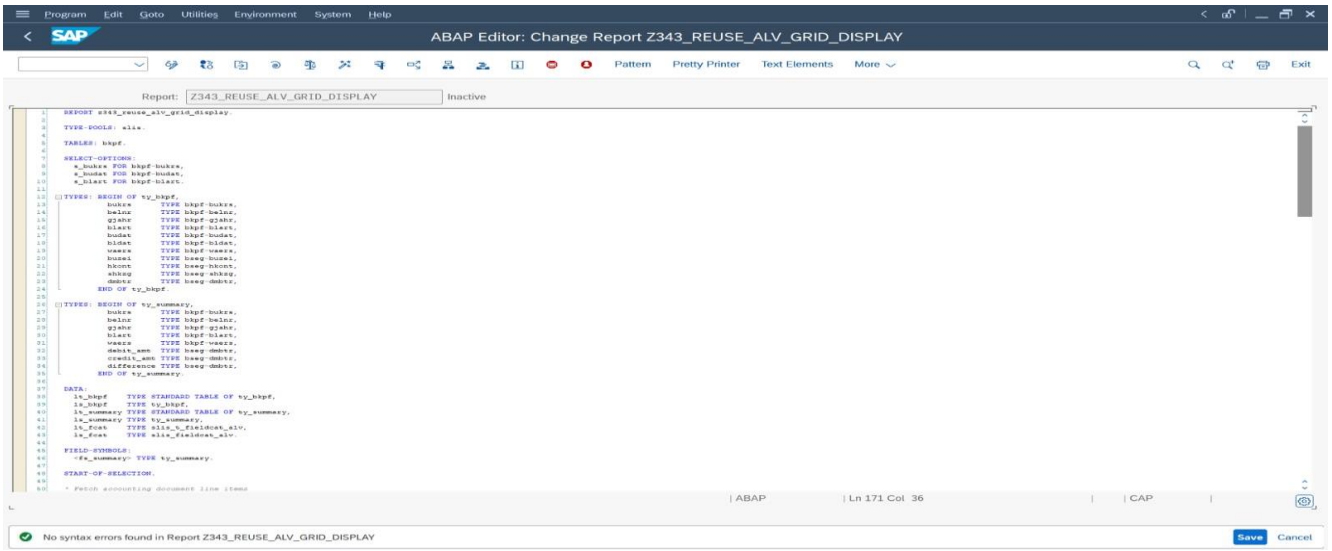


Figure 2: Syntax_Check_Activation:

Check:



Activate:

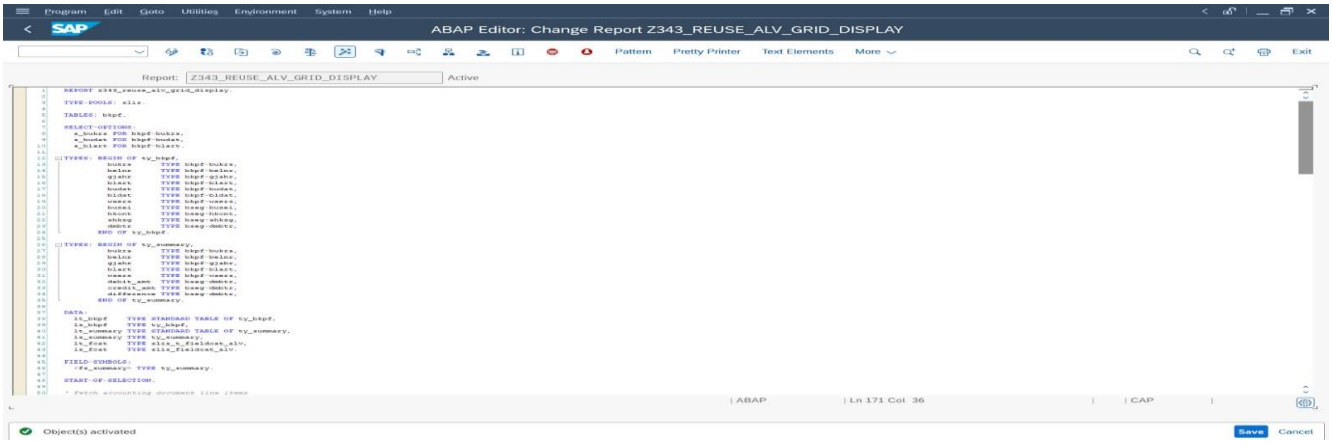
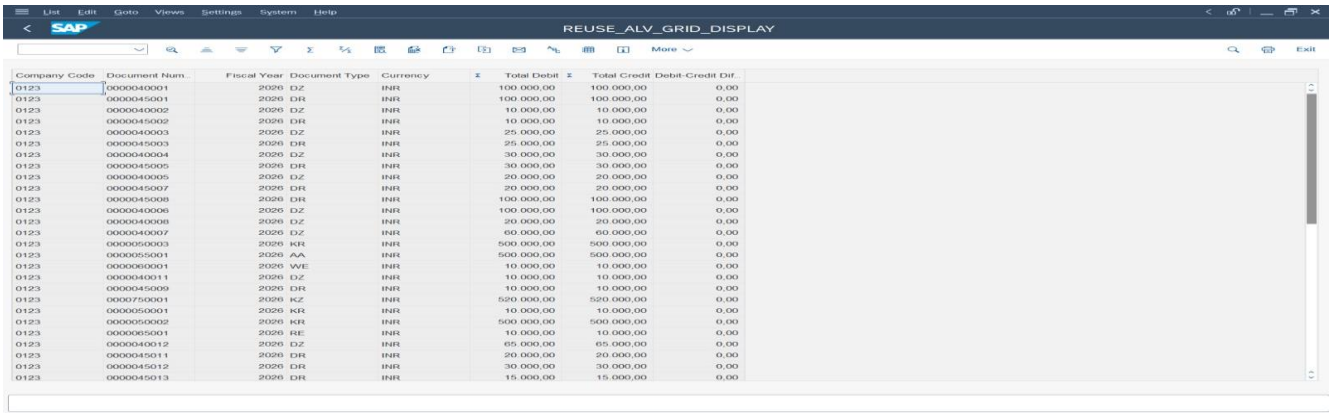
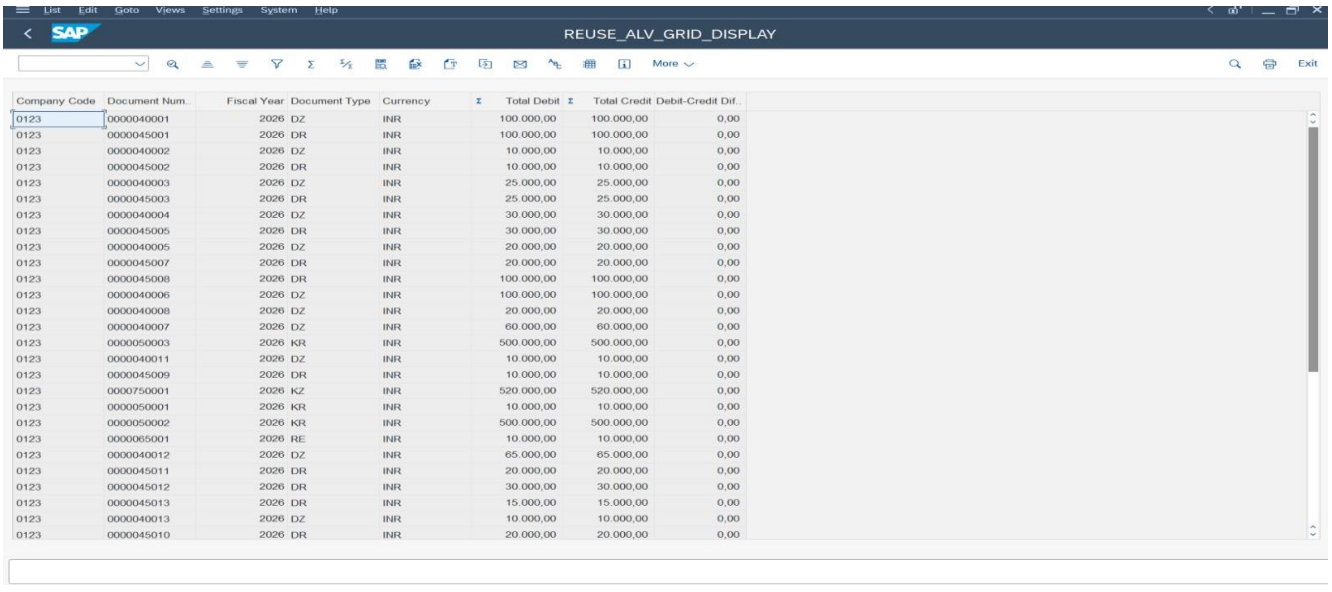


Figure 3: Final_ALV_Output:



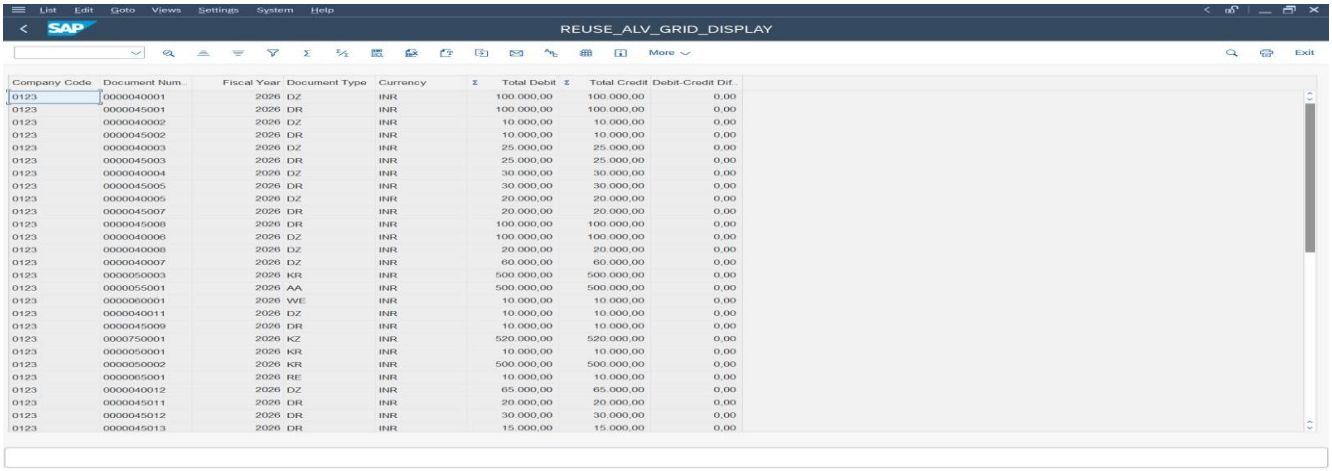
Company Code	Document Num...	Fiscal Year	Document Type	Currency	Total Debit	Total Credit	Debit-Credit Dif...
0123	0000040001	2026	DZ	INR	100.000,00	100.000,00	0,00
0123	0000045001	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000040002	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045002	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000040003	2026	DZ	INR	25.000,00	25.000,00	0,00
0123	0000045003	2026	DR	INR	25.000,00	25.000,00	0,00
0123	0000040004	2026	DZ	INR	30.000,00	30.000,00	0,00
0123	0000045005	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000040005	2026	DZ	INR	20.000,00	20.000,00	0,00
0123	0000045007	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045008	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000040006	2026	DZ	INR	100.000,00	100.000,00	0,00
0123	0000040008	2026	DZ	INR	20.000,00	20.000,00	0,00
0123	0000040007	2026	DZ	INR	60.000,00	60.000,00	0,00
0123	0000050003	2026	KR	INR	500.000,00	500.000,00	0,00
0123	0000055001	2026	AA	INR	500.000,00	500.000,00	0,00
0123	0000060001	2026	WE	INR	10.000,00	10.000,00	0,00
0123	0000040011	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045009	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000750001	2026	KZ	INR	520.000,00	520.000,00	0,00
0123	0000050001	2026	KR	INR	10.000,00	10.000,00	0,00
0123	0000050002	2026	KR	INR	500.000,00	500.000,00	0,00
0123	0000065001	2026	RE	INR	10.000,00	10.000,00	0,00
0123	0000040012	2026	DZ	INR	65.000,00	65.000,00	0,00
0123	0000045011	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045012	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000045013	2026	DR	INR	15.000,00	15.000,00	0,00

Figure 4: Debit_Credit_Validation :



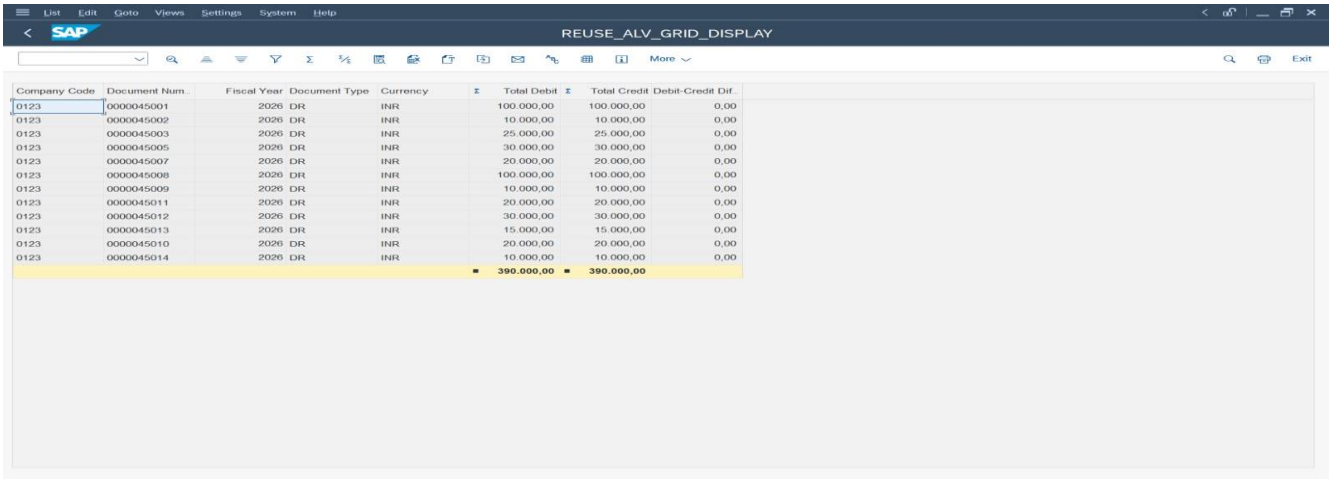
Company Code	Document Num...	Fiscal Year	Document Type	Currency	Total Debit	Total Credit	Debit-Credit Dif...
0123	0000040001	2026	DZ	INR	100.000,00	100.000,00	0,00
0123	0000045001	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000040002	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045002	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000040003	2026	DZ	INR	25.000,00	25.000,00	0,00
0123	0000045003	2026	DR	INR	25.000,00	25.000,00	0,00
0123	0000040004	2026	DZ	INR	30.000,00	30.000,00	0,00
0123	0000045005	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000040005	2026	DZ	INR	20.000,00	20.000,00	0,00
0123	0000045007	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045008	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000040006	2026	DZ	INR	100.000,00	100.000,00	0,00
0123	0000040008	2026	DZ	INR	20.000,00	20.000,00	0,00
0123	0000040007	2026	DZ	INR	60.000,00	60.000,00	0,00
0123	0000050003	2026	KR	INR	500.000,00	500.000,00	0,00
0123	0000040011	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045009	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000750001	2026	KZ	INR	520.000,00	520.000,00	0,00
0123	0000050001	2026	KR	INR	10.000,00	10.000,00	0,00
0123	0000050002	2026	KR	INR	500.000,00	500.000,00	0,00
0123	0000065001	2026	RE	INR	10.000,00	10.000,00	0,00
0123	0000040012	2026	DZ	INR	65.000,00	65.000,00	0,00
0123	0000045011	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045012	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000045013	2026	DR	INR	15.000,00	15.000,00	0,00
0123	0000040013	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045010	2026	DR	INR	20.000,00	20.000,00	0,00

Figure 5: Posting_Date_Filter_Test :



Company Code	Document Num...	Fiscal Year	Document Type	Currency	Total Debit	Total Credit	Debit-Credit Dif...
0123	0000040001	2026	DZ	INR	100.000,00	100.000,00	0,00
0123	0000045001	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000040002	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045002	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000040003	2026	DZ	INR	25.000,00	25.000,00	0,00
0123	0000045003	2026	DR	INR	25.000,00	25.000,00	0,00
0123	0000040004	2026	DZ	INR	30.000,00	30.000,00	0,00
0123	0000045005	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000040005	2026	DZ	INR	20.000,00	20.000,00	0,00
0123	0000045007	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045008	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000040006	2026	DZ	INR	100.000,00	100.000,00	0,00
0123	0000040008	2026	DZ	INR	20.000,00	20.000,00	0,00
0123	0000040007	2026	DZ	INR	60.000,00	60.000,00	0,00
0123	0000050003	2026	KR	INR	500.000,00	500.000,00	0,00
0123	0000055001	2026	AA	INR	500.000,00	500.000,00	0,00
0123	0000060001	2026	WE	INR	10.000,00	10.000,00	0,00
0123	0000040011	2026	DZ	INR	10.000,00	10.000,00	0,00
0123	0000045009	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000750001	2026	KZ	INR	520.000,00	520.000,00	0,00
0123	0000050001	2026	KR	INR	10.000,00	10.000,00	0,00
0123	0000050002	2026	KR	INR	500.000,00	500.000,00	0,00
0123	0000065001	2026	RE	INR	10.000,00	10.000,00	0,00
0123	0000040012	2026	DZ	INR	65.000,00	65.000,00	0,00
0123	0000045011	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045012	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000045013	2026	DR	INR	15.000,00	15.000,00	0,00

Figure 6: Document_Type_Filter_Test:



Company Code	Document Num...	Fiscal Year	Document Type	Currency	Total Debit	Total Credit	Debit-Credit Dif...
0123	0000045001	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000045002	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000045003	2026	DR	INR	25.000,00	25.000,00	0,00
0123	0000045005	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000045007	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045008	2026	DR	INR	100.000,00	100.000,00	0,00
0123	0000045009	2026	DR	INR	10.000,00	10.000,00	0,00
0123	0000045011	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045012	2026	DR	INR	30.000,00	30.000,00	0,00
0123	0000045013	2026	DR	INR	15.000,00	15.000,00	0,00
0123	0000045010	2026	DR	INR	20.000,00	20.000,00	0,00
0123	0000045014	2026	DR	INR	10.000,00	10.000,00	0,00
					390.000,00	**390.000,00**	

Figure 7: No_Data_Message_Test:

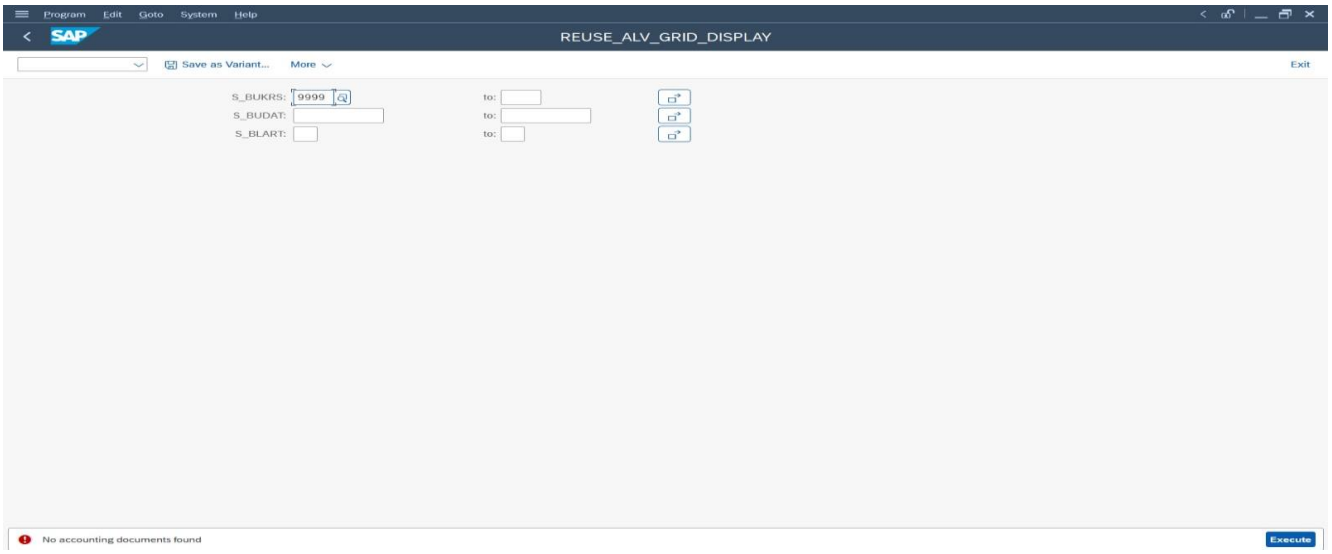


Figure 8: BSEG_Debit_Credit_Verification:

